

- Matrix $A_{m \times n}$ can be decomposed into the matrix product $U_{m \times m} \Sigma_{m \times n} V_{n \times n}^\dagger$.
- The columns of U will be eigenvectors of $AA^\dagger$ and U will be Orthogonal/Unitary ($U^\dagger U = I$).
- The columns of V will be eigenvectors of $A^\dagger A$ and V will be Orthogonal/Unitary ($V^\dagger V = I$).
- Σ will be a diagonal matrix consisting of singular values.

